

# Value Iteration

Instructor

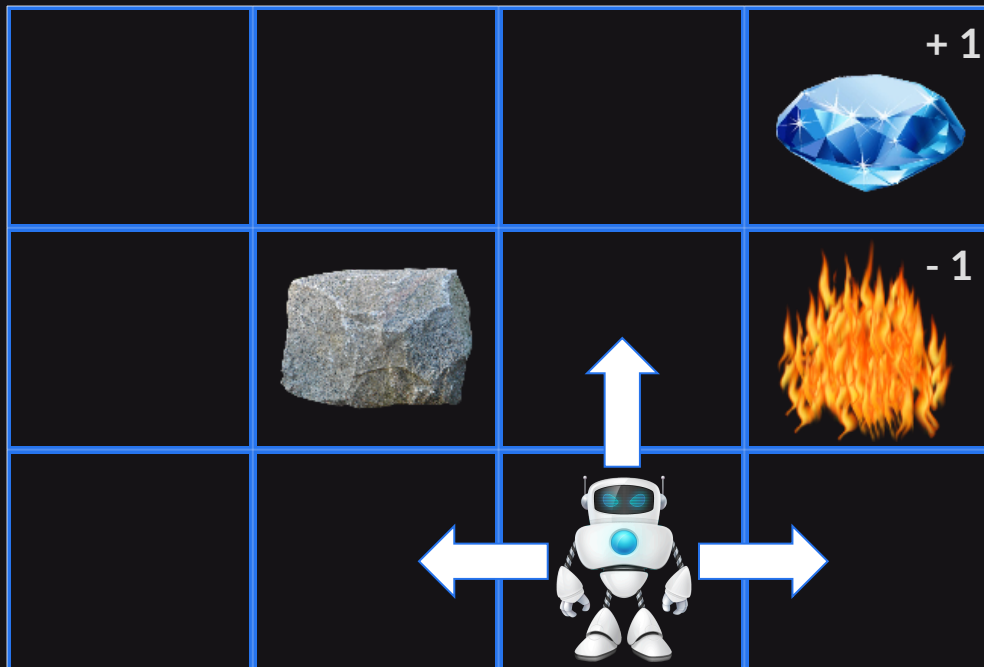
Dr. Mayank Baranwal

Senior Scientist, TCS Research



# Optimal Value Function $v_*$

$$v_*(s) = \max_{\pi} \mathbb{E}_{\pi} \left[ \sum_{k=0}^{T-1} \gamma^k R_{k+1} \mid S_0 = s, \pi \right]$$

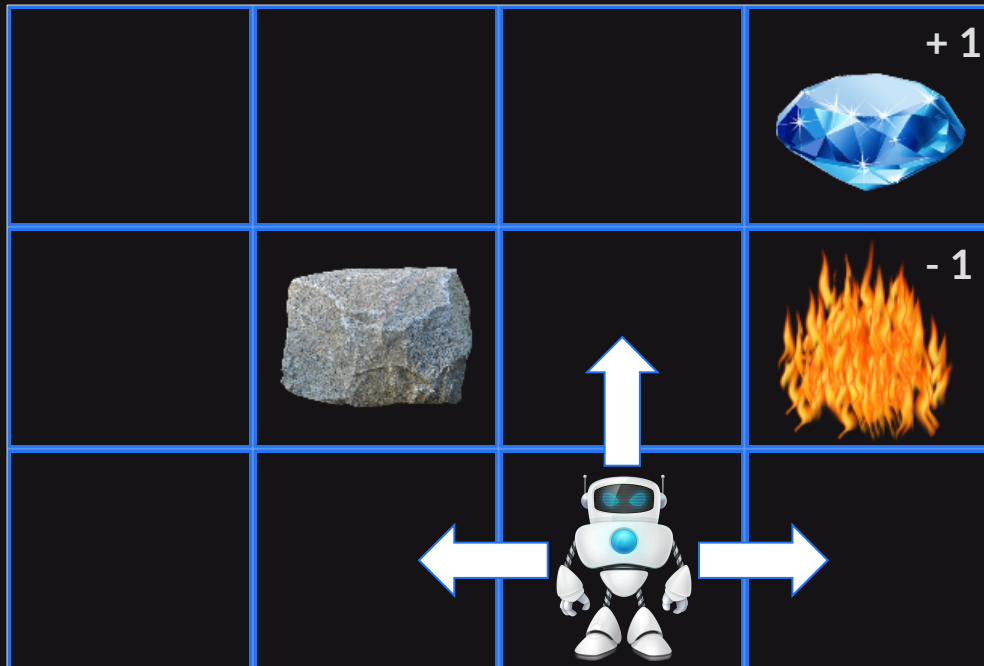


Assume  $T$  is sufficiently large, and  $\gamma = 1$

$$\begin{aligned} v_*(4, 3) &= 1 \\ v_*(3, 3) &= 1 \\ v_*(2, 3) &= 1 \\ v_*(1, 1) &= 1 \\ v_*(4, 2) &= -1 \end{aligned}$$

# Optimal Value Function $v_*$

$$v_*(s) = \max_{\pi} \mathbb{E}_{\pi} \left[ \sum_{k=0}^{T-1} \gamma^k R_{k+1} \mid S_0 = s, \pi \right]$$



Assume  $T$  is sufficiently large, and  $\gamma = 0.9$

$$v_*(4, 3) = 1$$

$$v_*(3, 3) = 0.9$$

$$v_*(2, 3) = 0.9^2 = 0.81$$

$$v_*(1, 1) = 0.9^5 = 0.59$$

$$v_*(4, 2) = -1$$

# Optimal Value Function $v_*$

$$v_*(s) = \max_{\pi} \mathbb{E}_{\pi} \left[ \sum_{k=0}^{T-1} \gamma^k R_{k+1} \mid S_0 = s, \pi \right]$$



Assume  $T$  is sufficiently large, and  $\gamma=0.9$ , and the probability of successful action is 0.8

$$\begin{aligned} v_*(4, 3) &= 1 \\ v_*(3, 3) &= 0.9 * (0.8 * v_*(4, 3) + 0.1 * v_*(3, 3) + 0.1 * v_*(2, 3)) \\ v_*(2, 3) &= \\ v_*(1, 1) &= \\ v_*(4, 2) &= \end{aligned}$$

# Value Iteration

Recall Bellman Optimality Equation for  $v_*$

$$v_*(s) = \max_a \sum_{s' \in S} P_{ss'}^a (R_{ss'}^a + \gamma v_*(s'))$$

Step 0

Initialize  $v_*^0(s) \leftarrow 0$ .

Step 1

$$v_*^1(s) \leftarrow \max_a \sum_{s' \in S} P_{ss'}^a (R_{ss'}^a + \gamma v_*^0(s')).$$

Bellman update/backup

Step k

$$v_*^k(s) \leftarrow \max_a \sum_{s' \in S} P_{ss'}^a (R_{ss'}^a + \gamma v_*^{k-1}(s')).$$

## Value Iteration

$$v_*^0(s) \leftarrow 0.$$

|      |                                                                                     |      |      |
|------|-------------------------------------------------------------------------------------|------|------|
| 0.00 | 0.00                                                                                | 0.00 | 0.00 |
| 0.00 |  | 0.00 | 0.00 |
| 0.00 | 0.00                                                                                | 0.00 | 0.00 |

## Value Iteration

$$v_*^1(s) \leftarrow \max_a \sum_{s' \in S} P_{ss'}^a (R_{ss'}^a + \gamma v_*^0(s')).$$

|      |                                                                                     |      |       |
|------|-------------------------------------------------------------------------------------|------|-------|
| 0.00 | 0.00                                                                                | 0.00 | 1.00  |
| 0.00 |  | 0.00 | -1.00 |
| 0.00 | 0.00                                                                                | 0.00 | 0.00  |

## Value Iteration

$$v_*^2(s) \leftarrow \max_a \sum_{s' \in S} P_{ss'}^a (R_{ss'}^a + \gamma v_*^1(s')).$$

|      |                                                                                     |      |       |
|------|-------------------------------------------------------------------------------------|------|-------|
| 0.00 | 0.00                                                                                | 0.72 | 1.00  |
| 0.00 |  | 0.00 | -1.00 |
| 0.00 | 0.00                                                                                | 0.00 | 0.00  |

## Value Iteration

$$v_*^4(s) \leftarrow \max_a \sum_{s' \in S} P_{ss'}^a (R_{ss'}^a + \gamma v_*^3(s')).$$

|      |                                                                                     |      |       |
|------|-------------------------------------------------------------------------------------|------|-------|
| 0.37 | 0.66                                                                                | 0.83 | 1.00  |
| 0.00 |  | 0.51 | -1.00 |
| 0.00 | 0.00                                                                                | 0.31 | 0.00  |

## Value Iteration

$$v_*^k(s) \leftarrow \max_a \sum_{s' \in S} P_{ss'}^a (R_{ss'}^a + \gamma v_*^{k-1}(s')).$$

|      |                                                                                     |      |       |
|------|-------------------------------------------------------------------------------------|------|-------|
| 0.64 | 0.74                                                                                | 0.85 | 1.00  |
| 0.57 |  | 0.57 | -1.00 |
| 0.49 | 0.43                                                                                | 0.48 | 0.28  |

*Repeat until convergence!*

# Thank You

---